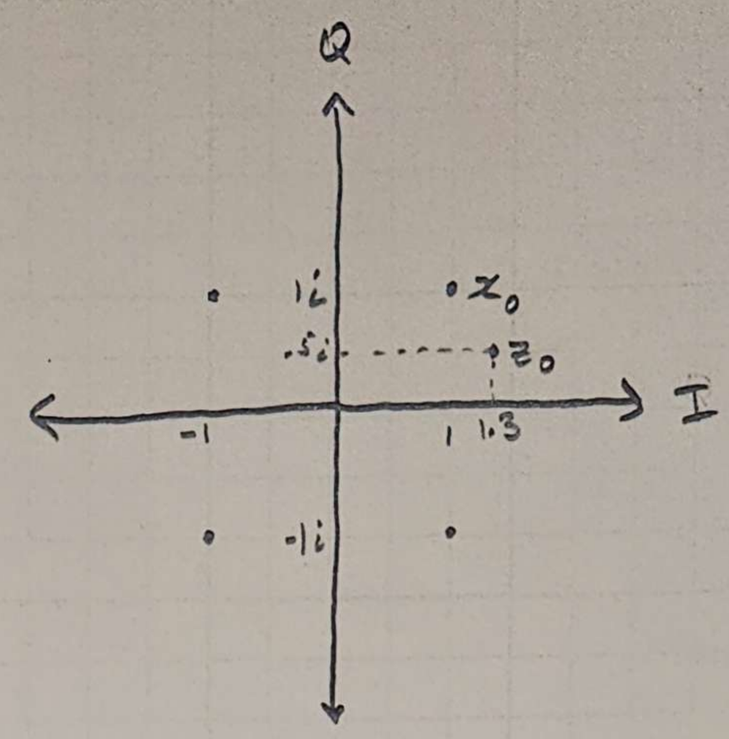
Problem 1 (p.1)	EE 417	HW4

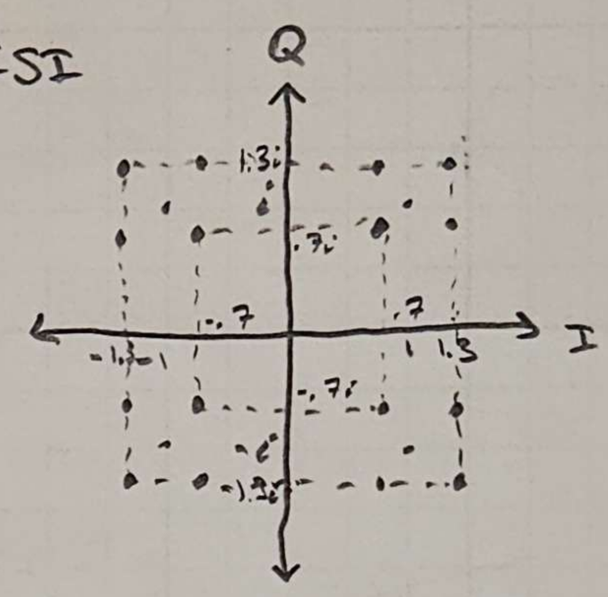
a) $z_k = x_k + n_k$



b) $z_k = x_k + .3x_{k-1} + n_k$

$\sigma_n^2 = 10^{-6}$, so basically no noise, only ISI

x_k	x_{k-1}	z_k
$1+i$	$1+i$	$1.3+1.3i$
...	$1-i$	$1.3+.7i$
	$-1+i$	$.7+1.3i$
	$-1-i$	$.7+.7i$
$1-i$	$1+i$	$1.3-.7i$
	$1-i$	$1.3-1.3i$
	$-1+i$	$.7-.7i$
	$-1-i$	$.7-1.3i$
$-1+i$	$1+i$	$-.7+1.3i$
	$1-i$	$-.7+.7i$
	$-1+i$	$-1.3+1.3i$
	$-1-i$	$-1.3+.7i$
$-1-i$	$1+i$	$-.7-.7i$
	$1-i$	$-.7-1.3i$
	$-1+i$	$-1.3-.7i$
	$-1-i$	$-1.3-1.3i$



c) $z_k = x_k + n_k$, $n_k \sim CN(1+j, .2)$

$\downarrow$

$z_k = x_k + 1 + j + \tilde{n}_k$, $\tilde{n}_k \sim (N(0, .2)$

$z_k - 1 - j - \tilde{n}_k = x_k$

$$x_k = \begin{cases} 1+j, & z_k^I > 1 \text{ AND } z_k^Q > i \\ 1-j, & z_k^I > 1 \text{ AND } z_k^Q < i \\ -1+j, & z_k^I < 1 \text{ AND } z_k^Q > i \\ -1-j, & z_k^I < 1 \text{ AND } z_k^Q < i \end{cases}$$

$E_x = \frac{4}{6}(3) = 2$

$SNR = \frac{E_x}{\sigma_n^2} = \frac{2}{.2} = 10$

$BER \le 2Q(\sqrt{10})$

$BER \le 2Q(3.16)$

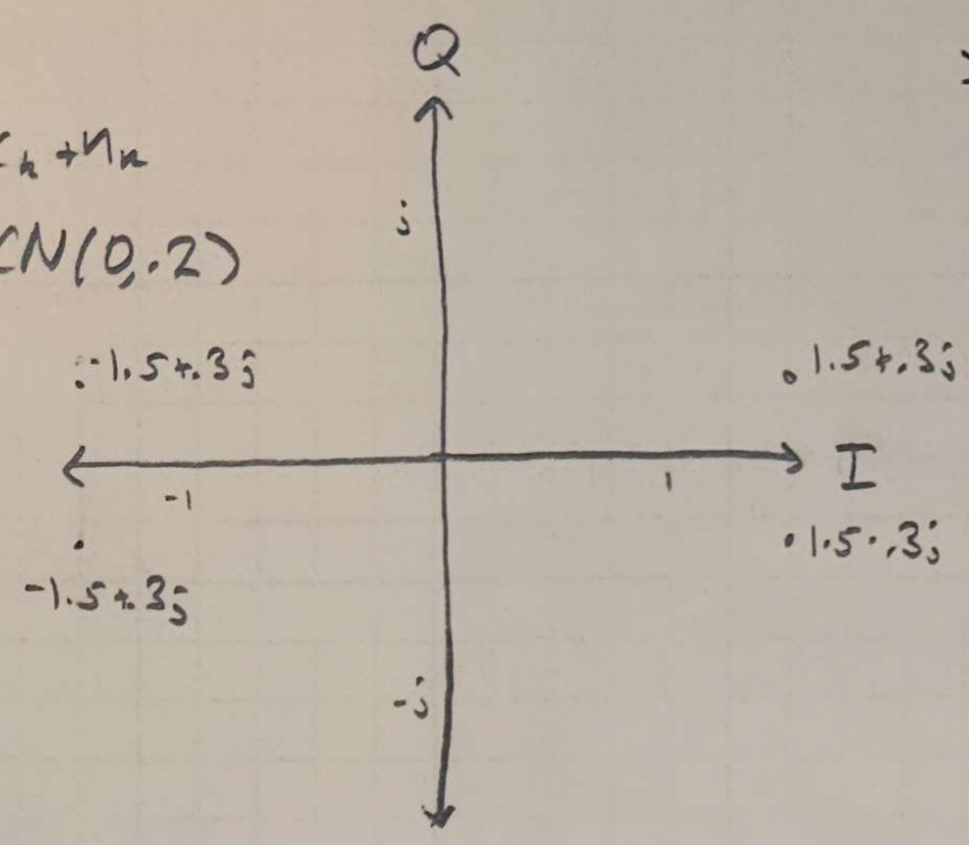
$\boxed{BER \le 7.83 \times 10^{-4}}$

d)

$Z_k = x_k + n_k$

$n_k \sim CN(0, .2)$

$$x_k = \begin{cases} 1.5+.3j,\ 1.5-.3j \\ -1.5+.3j\ \ -1.5-.3j \end{cases}$$



$\sigma_n^2 = .2 \quad \sigma_{n^Q}^2 = 0.1 = \sigma_{n^I}^2$

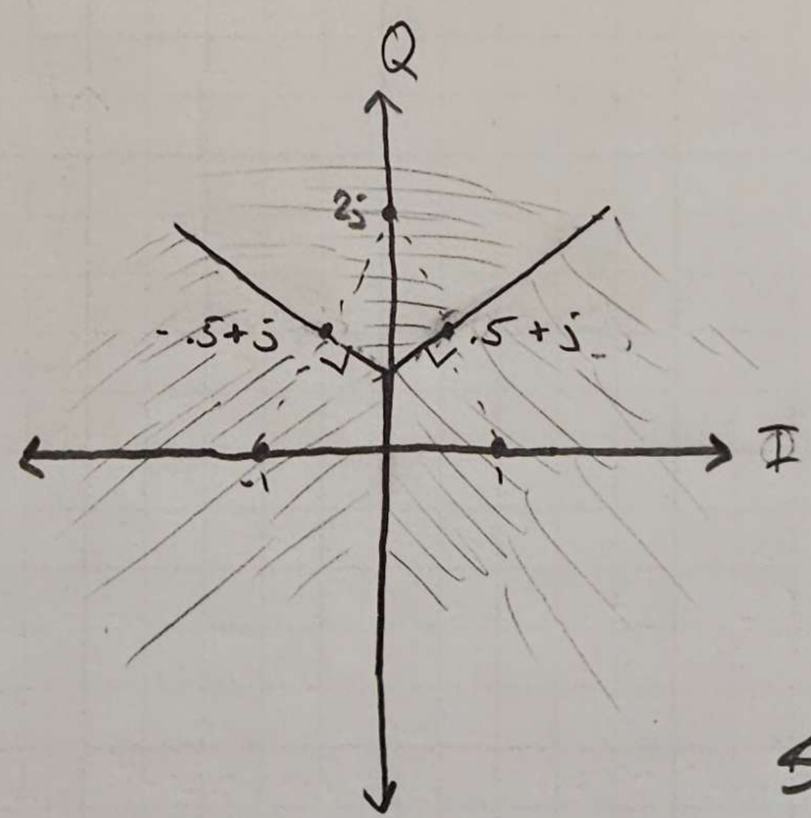
$d_{min} = .3 \cdot 2 = \boxed{0.6}$

$P_e \le P(n_k^I < -1.5) + P(n_k^Q < -0.3)$, ignore $P(n_k^I < \frac{-1.5}{\sigma_{n^I}})$ and $P(n_k^Q < \frac{-.3}{\sigma_{n^Q}})$

$\le Q(\frac{1.5}{\sqrt{.1}}) + Q(\frac{0.3}{\sqrt{.1}}) = 1.0507 \times 10^{-6} + 0.1714$

$\boxed{BER \le 0.1714}$

e)



$\boxed{d_{min} = 2}$

$\sqrt{1^2+2^2} = \sqrt{5}$

f)

$E_x = \frac{4+1+1}{3} = \boxed{2}$

$SNR = \frac{E_x}{\sigma_n^2} \rightarrow \sigma_n = \sqrt{\frac{2}{SNR}}$

$P_e \le \sum_{x \in S} P(\hat{x} \ne x_k \mid x_k = x) P(x_k = x) = \frac{2}{3}(P(n_k^I > 1) + P(n_k^Q > \frac{\sqrt{5}}{2}))$

$+ \frac{1}{3}(P(n_k^Q > \frac{\sqrt{5}}{2}) \cdot 2)$

$= \frac{2}{3}(Q(\frac{1}{\sigma_n}) + Q(\frac{\sqrt{5}}{2\sigma_n})) + \frac{2}{3}(Q(\frac{\sqrt{5}}{2\sigma_n}))$

$\boxed{P_e \le \frac{2}{3}(Q(\sqrt{\frac{SNR}{2}}) + 2Q(\sqrt{\frac{5\,SNR}{2}}))}$

Problem 2 (p.1)	EE 417	HW4

Assume $z_k = x_k + n_k$

a) $SNR = \frac{E_x}{\sigma_n^2} = 18dB = 63.1$, $P_e < 10^{-3}$

$$P_e < 10^{-3} = 2\,Q\left(\frac{d}{\sqrt{2}\,\sigma_n}\right)$$

$$E_x = \left(\frac{d}{2\sin(\frac{\pi}{M})}\right)^2 \longrightarrow \sqrt{E_x} = \frac{d}{2\sin(\frac{\pi}{M})}$$

$$d = 2\sin\left(\frac{\pi}{M}\right)\sqrt{E_x}$$

$$P_e < 10^{-3} = 2\,Q\left(\frac{2\sin(\frac{\pi}{M})}{\sqrt{2}}\left(\frac{\sqrt{E_x}}{\sigma_n}\right)\right) = 2\,Q\left(\frac{2\sin(\frac{\pi}{M})}{\sqrt{2}}\sqrt{SNR}\right)$$

$M = 18$, solved numerically

b) $B = 30\times10^3$ Hz $\qquad q(t) = \mathrm{sinc}\left(\frac{t}{T}\right)\frac{\cos(\pi\alpha t)T}{1-4\alpha^2t^2}$, $\alpha = .35$

$Q(f)$ has a width of $\frac{1+\alpha}{2T}$

$$\frac{1.35}{2T} = 30\times10^3\,Hz \longrightarrow T = 2.25\times10^{-5}\,s$$

$$\frac{1}{T} = 4.44\times10^4\,Hz$$

$$\boxed{1.33\times10^5\ \frac{\text{bits}}{\text{second}}}$$

c) $b=3$

$P_e = 10^{-3}$ $M=8$

$$\frac{10^{-3}}{2} = Q\left(\frac{2\sin(\frac{\pi}{8})}{\sqrt{2}}\sqrt{SNR}\right) \longrightarrow SNR \approx 37 = 15.7\,dB$$

$b=4$, $M=16$

$$\frac{10^{-3}}{2} = Q\left(\frac{2\sin(\frac{\pi}{16})}{\sqrt{2}}\sqrt{SNR}\right) \longrightarrow SNR \approx 144 = 21.6\,dB$$

$b=5$, $M=32$

$$\frac{10^{-3}}{2} = Q\left(\frac{2\sin(\frac{\pi}{32})}{\sqrt{2}}\sqrt{SNR}\right) \longrightarrow SNR \approx 563 = 27.5\,dB$$

The trend is that SNR increases by 5.9 dB for every additional bit.

This makes sense because by increasing the bit count, you decrease minimum distance by a factor of 2.

And noise must also relatively decrease by a factor of 2 to maintain P_e. So this is an exponential relation, which is why dB increases linearly.

d) $P_e = 10^{-6}$, $b=3$

$$\frac{10^{-6}}{2} = Q\left(\frac{2\sin(\frac{\pi}{8})}{\sqrt{2}}\sqrt{SNR}\right) \longrightarrow SNR \approx 82 = \boxed{19.1\,dB}$$

e) $\frac{1}{d^p}$, $p=4$, $P_e = 10^{-6}$, $P = .1W$

$\downarrow$

$$SNR = 82 = \frac{E_{bc}}{\sigma_n^2} \cdot \frac{1}{d^4}$$

$$10^{1.8} = 63.1 = \frac{E_{bc}}{\sigma_n^2 \cdot d^4} = \frac{E_{bc}}{\sigma_n^2}\left(\frac{1}{1.2^4}\right) = \frac{E_x}{\sigma_n^2}(2.0736)$$

$$\frac{E_{bc}}{\sigma_n^2} = 30.428$$

$$82 = \frac{E_x}{\sigma_n^2} \cdot \frac{1}{d^4} = 30.428 \cdot \frac{1}{d^4} \rightarrow 2.695 = \frac{1}{d^4}$$

$\downarrow$

$$d = \sqrt[4]{\frac{1}{2.695}}$$

$$\boxed{d = 0.7805 \text{ km}}$$

f) $\frac{40\mu s}{2} = 20 \times 10^{-6} s$

$3 \times 10^8 m/s \cdot 20 \times 10^{-6} s = 6000 m =$ radius of 6 km.

$A \approx \pi r^2 = \pi (6 \times 10^3 m)^2 = 1.13 \times 10^8 m^2$

g)

$P_e = 10^{-3}$ Square: $\frac{10^{-3}}{4} = \left(1 - \frac{1}{\sqrt{M}}\right) Q\left(\sqrt{\frac{3 SNR}{M-1}}\right)$

$SNR = 18 dB = 63.1$

$M = 16$, $b = 4$

Bit rate is able to be $\frac{1}{3}$ times higher using M-QAM instead of PSK.

Problem 2 (p.4)	EE 417	HW4

h) $P_e = 10^{-3}$, $M = 16$, $b = 4$

$$\frac{10^{-3}}{4} \le \left(1 - \frac{1}{4}\right) Q\left(\sqrt{\frac{3\,SNR}{15}}\right)$$

$$\frac{10^{-3}}{3} \le Q\left(\sqrt{\frac{SNR}{5}}\right)$$

$$SNR = 58 = \boxed{17.6\,dB = SNR}$$

$b = 5$

$M = 32$

$$\frac{10^{-3}}{4} \le \left(1 - \frac{1}{\sqrt{2M}}\right) Q\left(\sqrt{\frac{3\,SNR}{\frac{31}{32}M - 1}}\right)$$

$$\frac{10^{-3}}{4} \le \frac{7}{8} Q\left(\sqrt{\frac{3\,SNR}{30}}\right)$$

$$\frac{2 \cdot 10^{-3}}{7} \le Q\left(\sqrt{\frac{SNR}{10}}\right) \rightarrow \boxed{SNR \approx 119 = 20.8\,dB}$$

$b = 6$

$M = 64$

$$\frac{2 \cdot 10^{-3}}{7} \le Q\left(\sqrt{\frac{3\,SNR}{63}}\right)$$

$$\frac{2 \cdot 10^{-3}}{7} \le Q\left(\sqrt{\frac{SNR}{21}}\right) \rightarrow \boxed{SNR \approx 250 = 24\,dB}$$

The trend is SNR increases by about 3.2 dB every additional bit added. This makes sense because every additional 1 bit decreases minimum distance by 2. So noise must correspondingly decrease by a factor of 2 to compensate. This is exponential, explaining why dB increases linearly.

Problem 2 (p.5)	EE 417	HW4

i) Spectral Efficiency for part b:

$$\frac{1.33 \times 10^5 \text{ bits/second}}{30 \times 10^5 \text{ Hz}} = \boxed{0.0443 \text{ bits}}$$

Part g:

$$\frac{1.77 \times 10^5 \text{ bits/second}}{30 \times 10^5 \text{ Hz}} = \boxed{0.0591 \text{ bits}}$$

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